

Environmental hazards

Environmental hazards

- **A hazard** is any agent that can cause harm or damage to humans, property, or the environment.
- **A risk** is the probability of suffering harm from a hazard that can cause injury, disease, death, economic loss, or damage
- **Risk assessment** is the process of using statistical methods to estimate how much harm a particular hazard can cause to human health or to the environment.
- **Risk management** involves deciding whether or how to reduce a particular risk to a certain level and at what cost.

Environmental hazards

We face different types of environmental hazards

- biological hazards
 - chemical hazards
 - nuclear hazards
 - natural hazards
 - cultural hazards
 - life style hazards
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- **Physical hazards** such as fire, earthquakes, volcanic eruptions, floods, and storms.
 - **Cultural hazards** such as unsafe working conditions, criminal assault, poverty, etc.
 - **Lifestyle hazards** such as smoking, making poor food choices, drinking too much alcohol, etc.

Environmental hazards

Biological hazards

- infectious disease
- transmissible disease
- non-transmissible disease

Environmental hazards

Biological hazards

Infectious disease: Infectious disease are disease caused by microorganisms.

An infectious disease is caused when a pathogen such as a bacterium, virus, or parasite invades the body and multiplies in its cells and tissues.

examples: flu, malaria, tuberculosis, and measles.

A **bacterial** disease such as tuberculosis spreads as the bacteria multiply.

A **viral** disease such as flu or HIV spreads as viruses take over a cell's genetic mechanisms to copy themselves

Environmental hazards

Biological hazards

- **transmissible disease** (also called a contagious or communicable disease) is an infectious disease that can be transmitted from one person to another.

Examples are flu, tuberculosis, and measles.

- **non-transmissible disease** is caused by something other than a living organism and does not spread from one person to another.

Examples: cardiovascular (heart and blood vessel) diseases, most cancers, asthma, diabetes, and malnutrition.

Environmental hazards

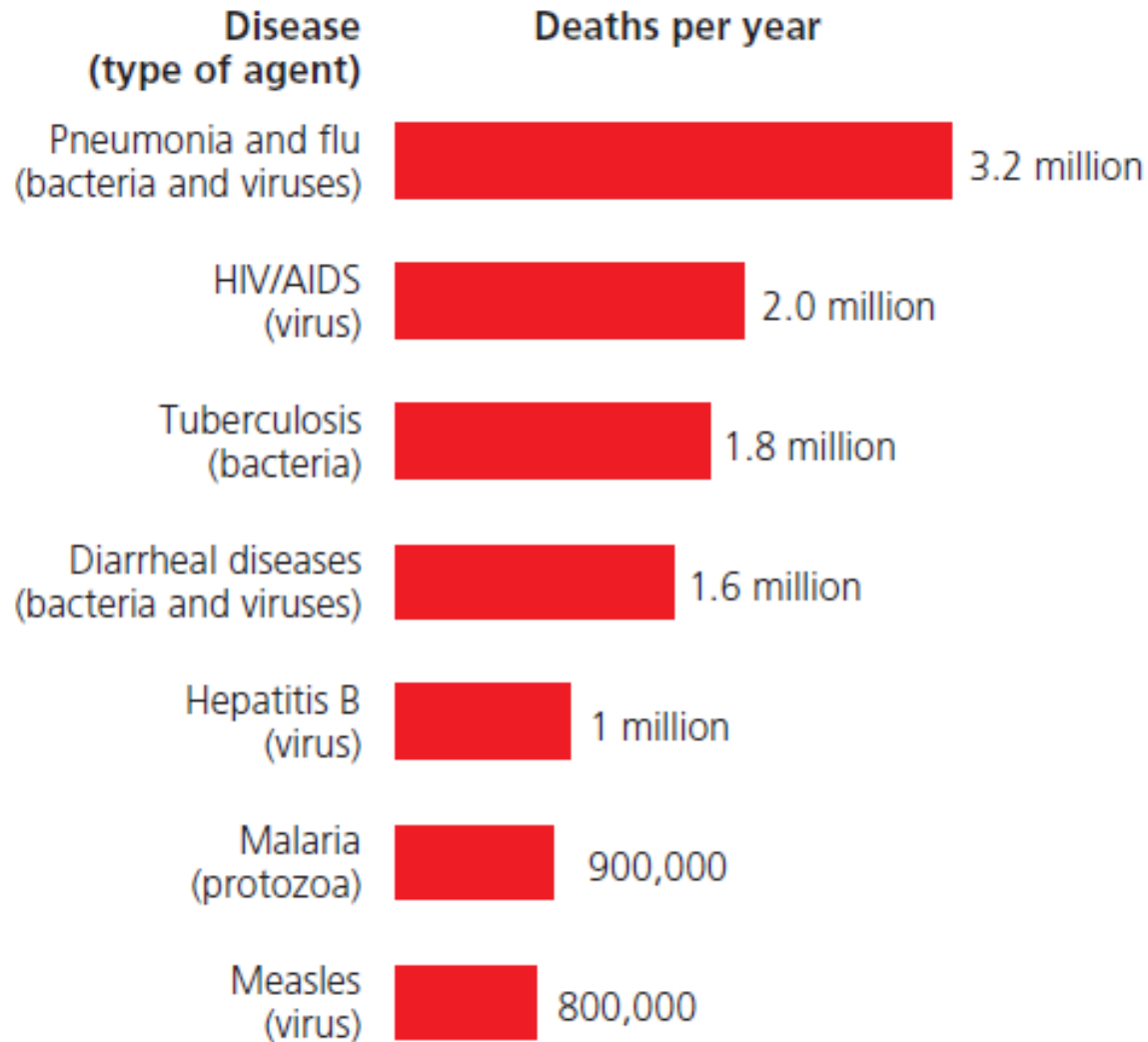
Biological hazards- Infectious Diseases Are Still Major Health Threats

- Infectious diseases remain as serious health threats, especially in less-developed countries.
- A large-scale outbreak of an infectious disease in an area or a country is called an *epidemic*.
- A global epidemic such as tuberculosis or AIDS is called a *pandemic*.
- One reason why infectious disease is still a serious threat is that many disease-carrying *bacteria have developed genetic resistance to widely used antibiotics*

Example: mosquitoes have become immune to widely used pesticides such as DDT

Environmental hazards

Biological hazards- Infectious Diseases Are Still Major Health Threats



Chemical Hazards

Chemical Hazards

A **toxic chemical** is one that can cause temporary or permanent harm or death to humans and animals.

In 2004, the EPA listed

- arsenic
- lead
- mercury
- vinyl chloride (used to make PVC plastics),
- polychlorinated biphenyls (PCBs)

as the top **five toxic substances** in terms of human and environmental health.

Chemical Hazards

A **toxic chemical** is one that can cause temporary or permanent harm or death to humans and animals.

There are three major types of potentially toxic agents.

- Carcinogens
- mutagens
- teratogens

Chemical Hazards

toxic chemical-carcinogens

Carcinogens are chemicals, types of radiation, or certain viruses that can cause or promote cancer.

Cancer is disease in which malignant cells multiply uncontrollably and create tumors that can damage the body and often lead to premature death.

Examples of carcinogens are arsenic, benzene, formaldehyde, gamma radiation, PCBs, radon, certain chemicals in tobacco smoke, ultraviolet(UV) radiation, vinyl chloride.

Typically, 10-40 years may elapse between the initial exposure to a carcinogen and the appearance of detectable cancer symptoms.

Partly because of this time lag, many healthy teenagers and young adults have trouble believing that their smoking, drinking, eating, and other habits today could lead to some form of cancer before they reach age 50.

Chemical Hazards

Toxic chemical-mutagens

Mutagens are chemicals or forms of radiation that cause or increase the frequency of mutations, or changes, in the DNA molecules found in cells.

Most mutations cause no harm but some can lead to cancers and other disorders.

For example,

- nitrous acid (HNO_2), formed by the digestion of nitrite (NO_2^-) preservatives in foods, can cause mutations linked to increases in stomach cancer in people who consume large amounts of processed foods and wine containing such preservatives.

3. Teratogens are chemicals that cause **harm or birth defects** to a foetus or embryo.

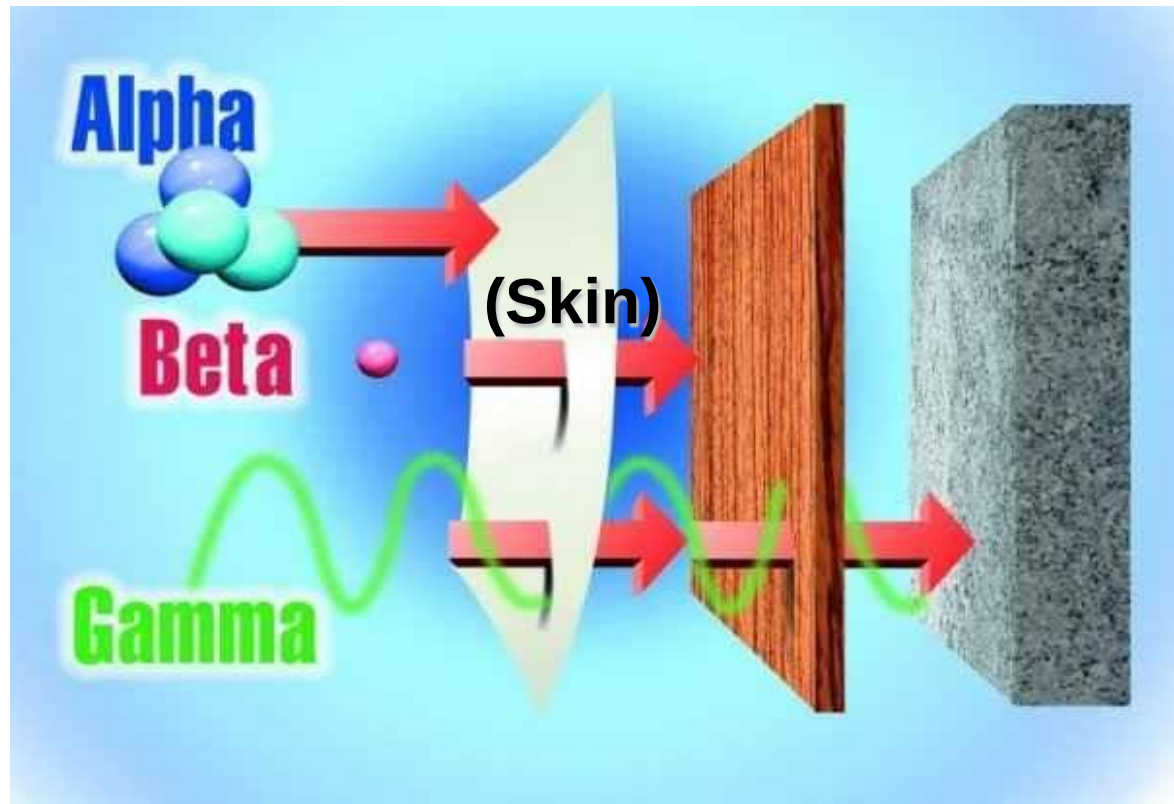
Ethyl alcohol is a teratogen. Drinking during pregnancy can lead to offspring with low birth weight and a number of physical, developmental, behavioural, and mental problems.

Other teratogens are angel dust, benzene, cadmium, formaldehyde, lead, mercury, mescaline, PCBs, phthalates, thalidomide, and vinyl chloride.

Nuclear Hazards: sources of radiation

- Radioactive substances undergo radioactive decay - unstable atoms to stable atoms.
- Radioactive decay release **energy** along with **Gamma rays** (high energy electromagnetic radiation), **Beta particles** (fast moving, -vely charged), and **Alpha particles** (fast moving, +vely charged).

Penetration
power of ionizing
radiations



Sources of Nuclear Radiation

- Nuclear radiation is the major cause of nuclear hazards, it can be of two types:

(1) Natural sources:

- > Cosmic rays from outer space , radioactive radon-222, soil, rocks, air, water and food which contain one or more radioactive substances .

(2) Anthropogenic sources

- > These sources are nuclear power plants , nuclear accidents , X-rays , diagnostic kits , test laboratories etc., where radioactive substances are used.

Effects of Nuclear Hazards to the environment

> In general, the ill effects produced by nuclear weapons are of two types:

- (i) Somatic
- (ii) Genetic

(1) Somatic effects:

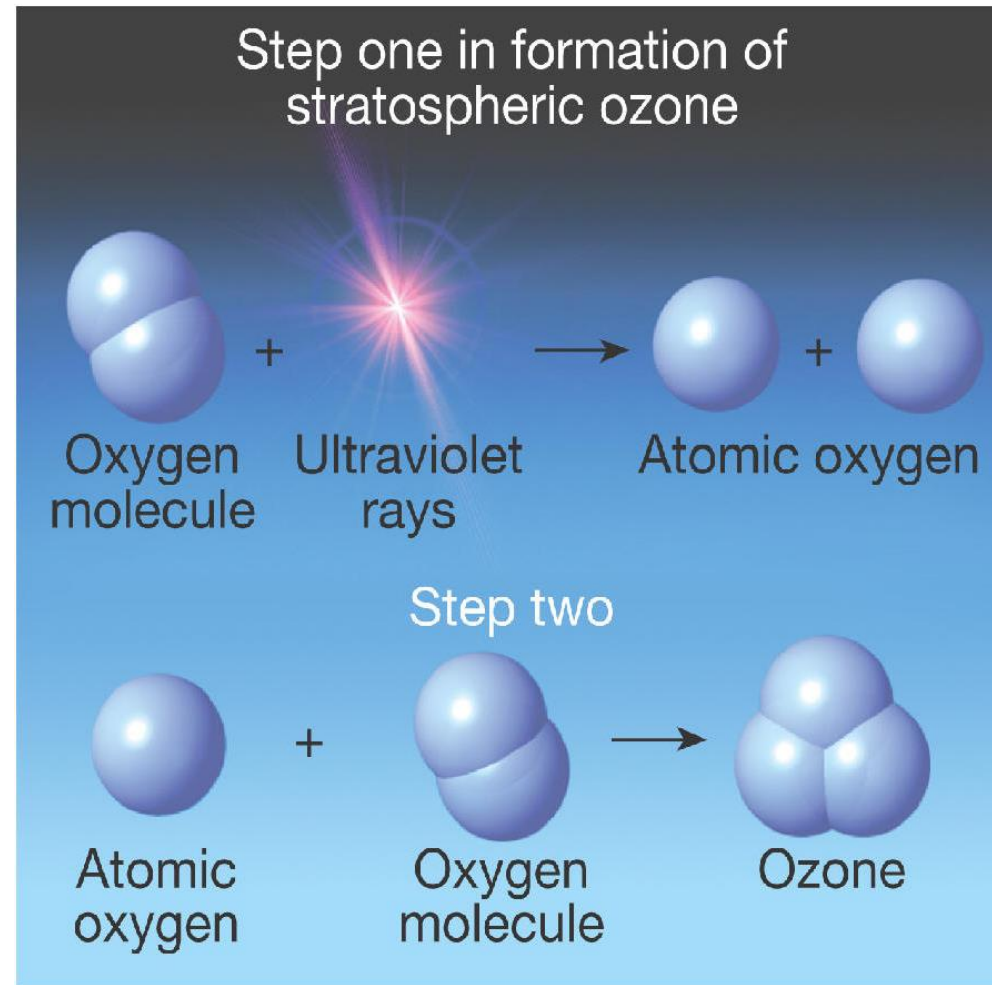
The general body cells of the person exposed to nuclear radiation are affected. This may lead to deadly diseases like cancer and shorten the lives of the people.

(2) Genetic effects:

The genes of the person exposed to nuclear radiation get damaged. The genetic disorders are passed onto the next generations. These effects are more dangerous than the somatic effects.

Formation and Breakdown of Stratospheric Ozone

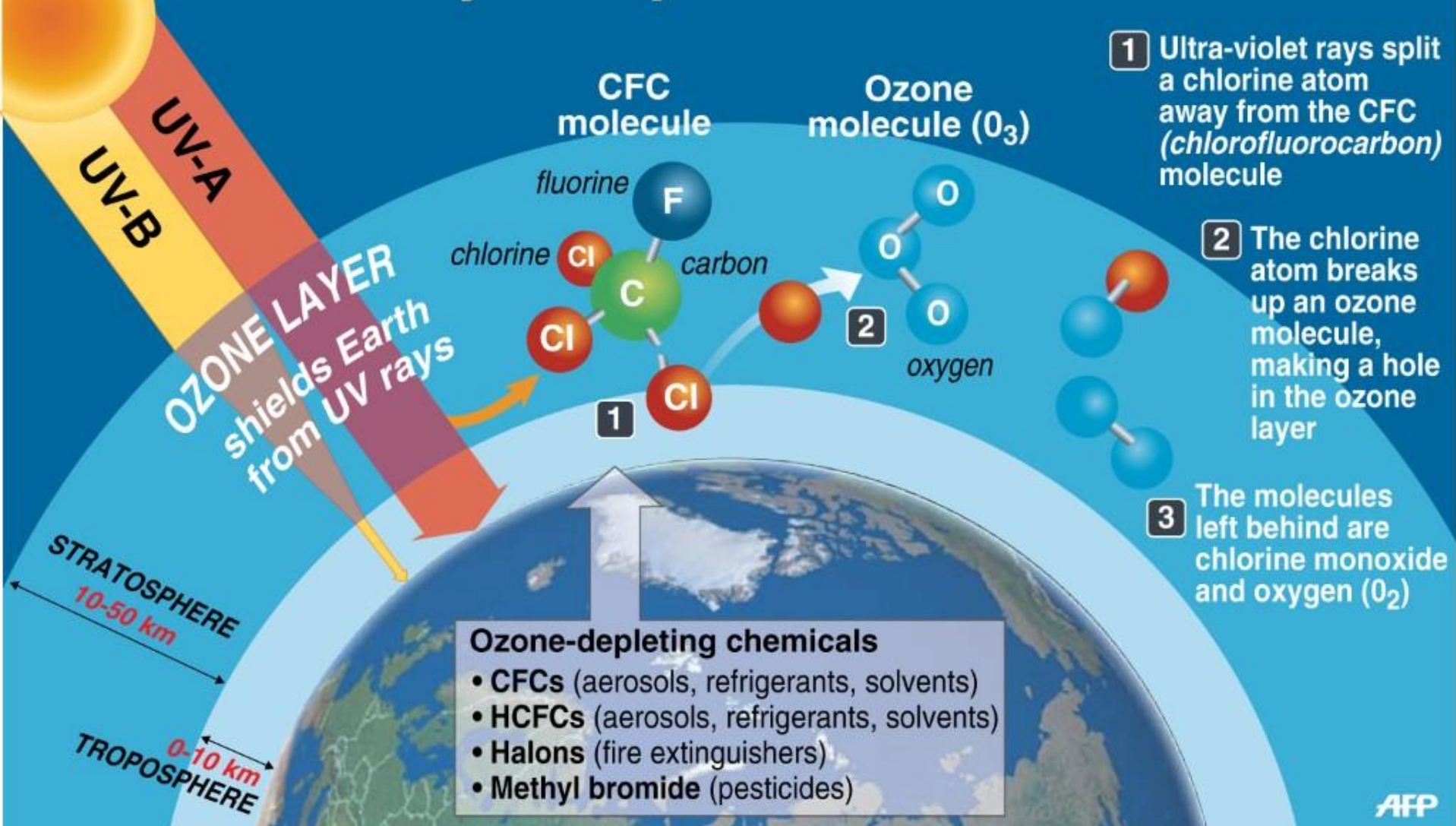
- Stratospheric ozone forms and breaks down naturally in a ***closed-loop cycle***
- UV-C radiation breaks the bond, that holds the oxygen molecule together, leaving ***two free oxygen atoms***
- The free oxygen atoms ***react with oxygen molecule*** to form ozone
- Ozone is broken down into oxygen molecule when it absorbs both UV-C and UV-B light. This free oxygen again may react with O_2 to form ozone. ***Thus ozone is formed and broken down in a cycle.***



Anthropogenic Contribution to Ozone Destruction

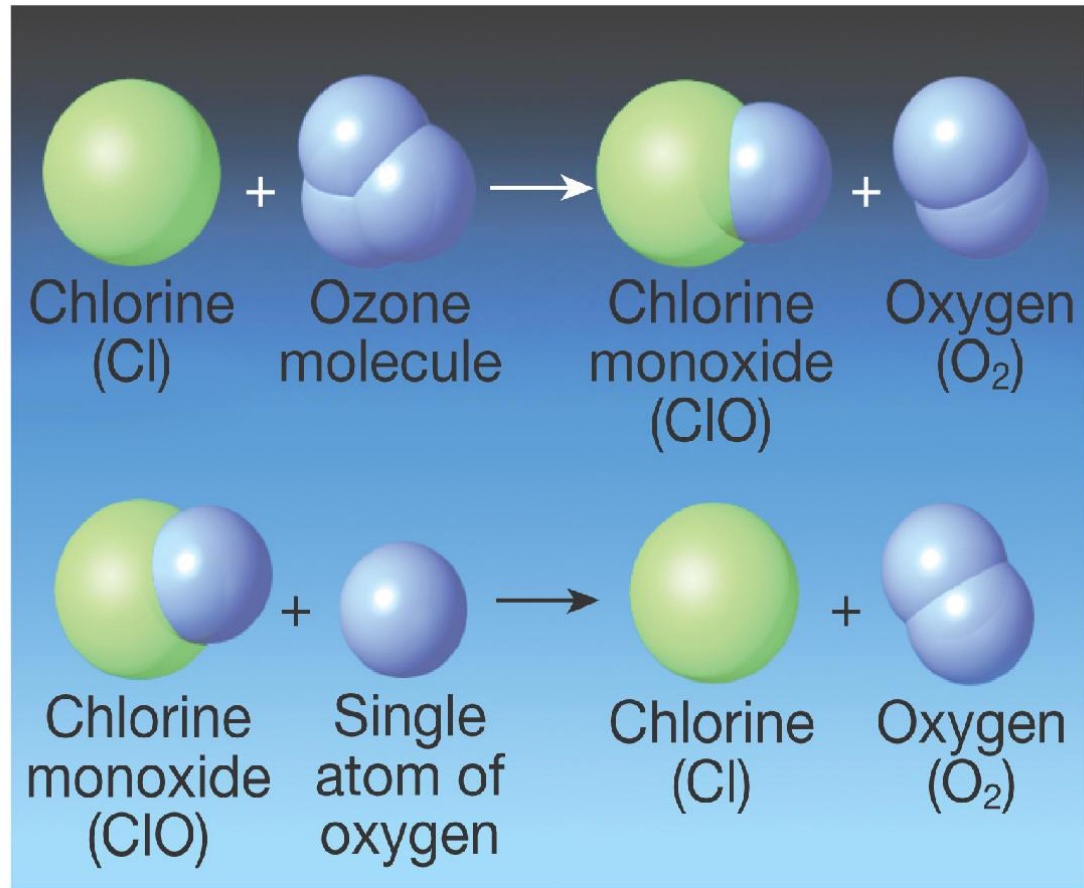
- Certain chemical catalysts can break down ozone, the most important one is **chlorine**.
- Major source of chlorine in stratosphere is a class of anthropogenic compounds known as **chlorofluorocarbons (CFCs)**.
- CFCs are the family of organic compounds whose properties make them ideal for use in **refrigeration and air conditioning**, as **propellant in aerosol cans to deliver ingredients such as deodorant and insect repellent**, and as **blowing agents to inject air into foam products like Styrofoam cups and foam insulation**.
- CFCs **does not degrade**, dissolve in water or undergo any significant chemical change, but slowly circulates in the atmosphere.
- However, **in presence of UV radiation** in stratosphere, the chlorine bond in CFC breaks

Ozone layer depletion



Chemistry of Ozone Depletion

- UV radiation hits CFC molecule and chlorine atom breaks away.
- Chlorine atom hits the ozone molecule and takes one oxygen atom to create **chlorine monoxide**, leaves one molecule of oxygen.

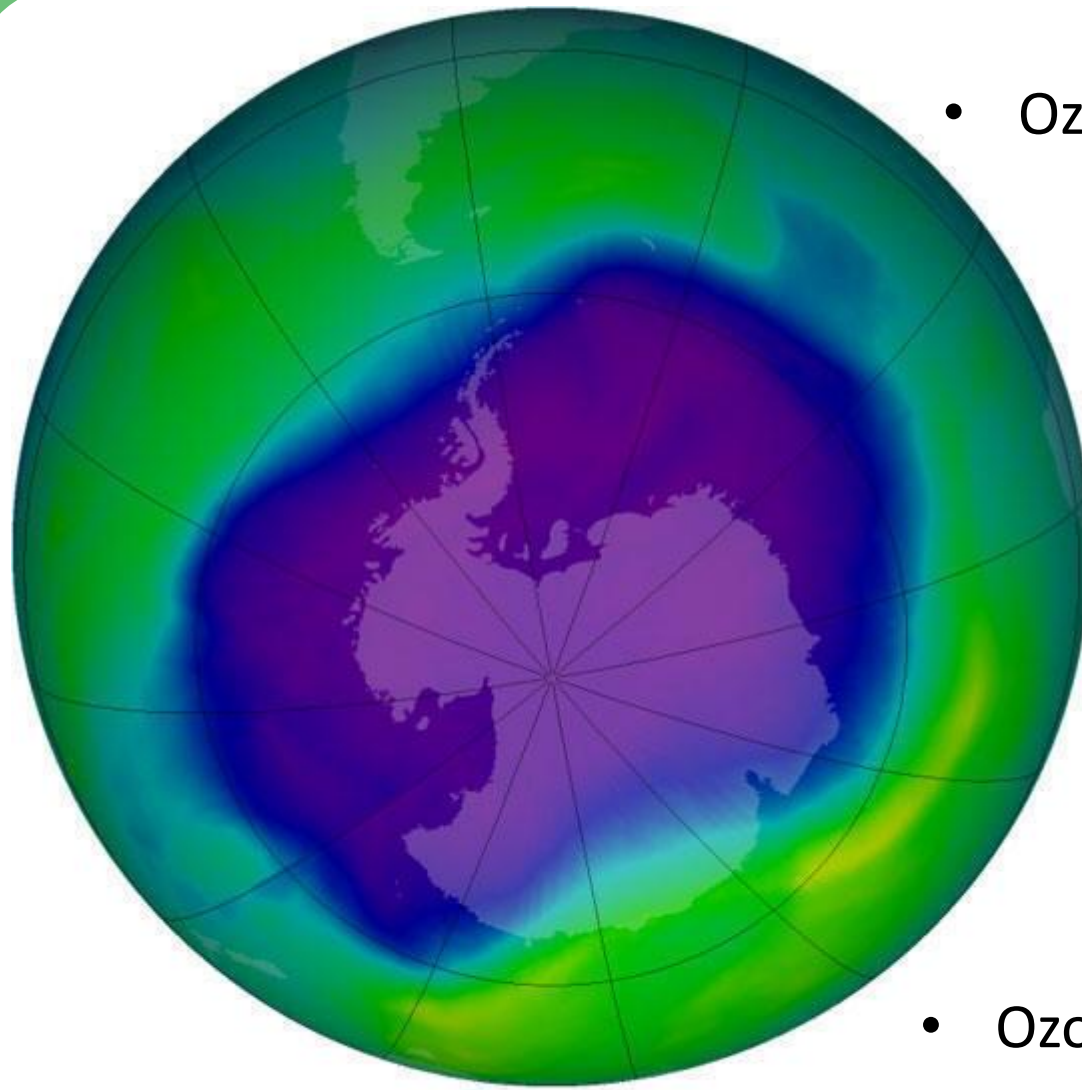


- Oxygen atom hits chlorine monoxide molecule and forms oxygen molecule and chlorine atom and this chlorine atom **repeats the depletion process**.

Depletion of Ozone Layer

- The stratospheric ozone in Antarctica had been *decreasing each year*, from 1979.
- It had also been observed that, in the Antarctica, ozone depletion was seasonal: each year depletion occurred from August through November.
- The depletion caused an area of severely reduced ozone concentrations over most of Antarctica, called as “*Ozone hole*”
- Decreased stratospheric ozone has increased the amount of UV-B radiation that reaches the earth's surface

Ozone “Hole” over Antarctica



Total Ozone (Dobson Units)

Color	Value (Dobson Units)
Purple	110
Dark Blue	220
Light Blue	330
Green	440
Yellow	550

- Ozone “hole” is not a hole at all.
- Ozone is *spread thinly throughout stratosphere in low quantities.*
- Ozone “hole” actually means the region where the ozone is *much diluted.*

Health Consequences of Ozone Depletion

- *Skin cancers, Sunburn, eye disorders, cataracts*
- *Premature aging of the skin*
- *Immune system damage*
- *DNA mutation of existing disease bacteria and virus*
- *Phytoplanktons are sensitive to ozone depletion. Ozone depletion will result in decrease in their population thereby affecting the population of zooplankton, fish, marine animals.*
- *Yield of vital crops like corn, rice, soybean, wheat will decrease.*
- *Degradation of paints, plastics and other polymer materials will result in economic loss due to the effects of ozone depletion.*

Greenhouse Effect

- Differences in the types of radiation emitted by the sun and earth, in combination with the greenhouse effect *causes the planet to warm*
- The greenhouse effect gets its name from the gardener's greenhouse
- Actual greenhouses warmed by the sun involves glass windows holding in heat, whereas the processes by which earth is warmed involves greenhouse gases radiating infrared energy back to earth

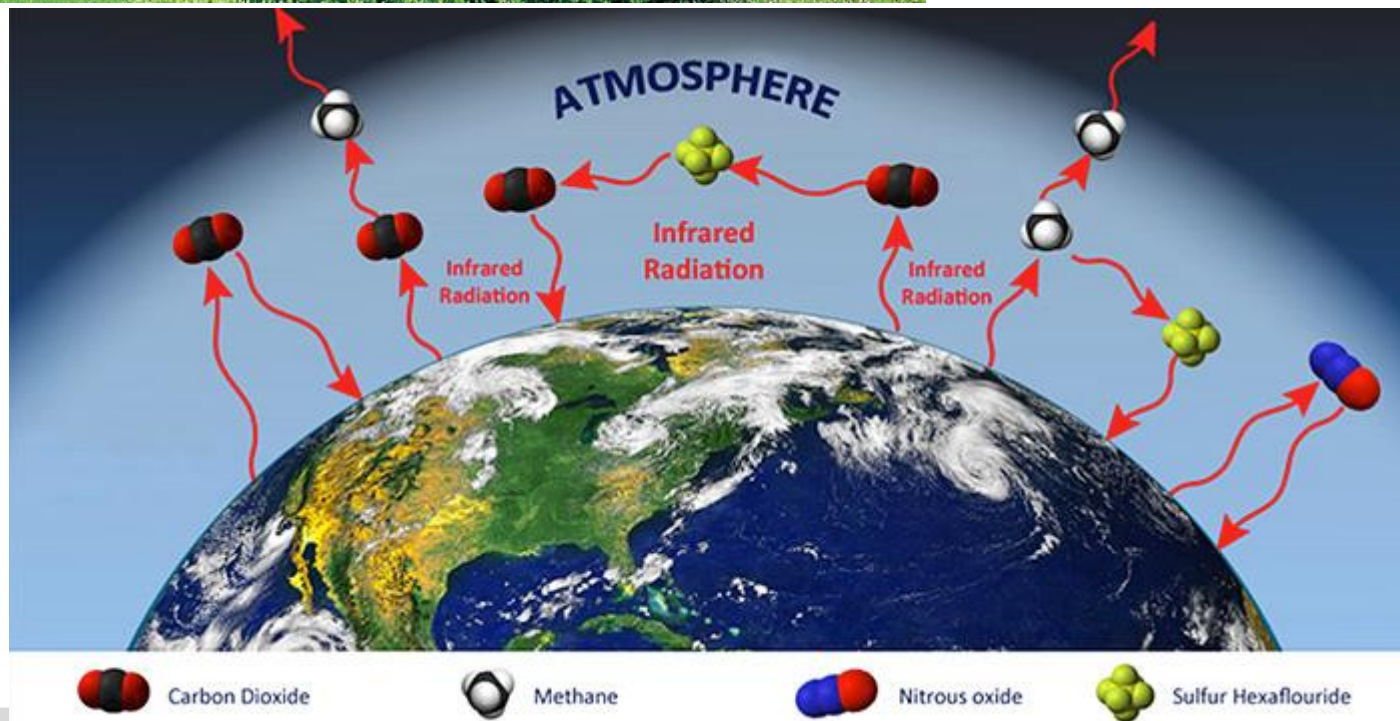


Gardener's Greenhouse

Made up of Glass windows

Earth's Greenhouse

Made up of Greenhouse Gases



① Incoming solar radiation consists primarily of UV and visible light.

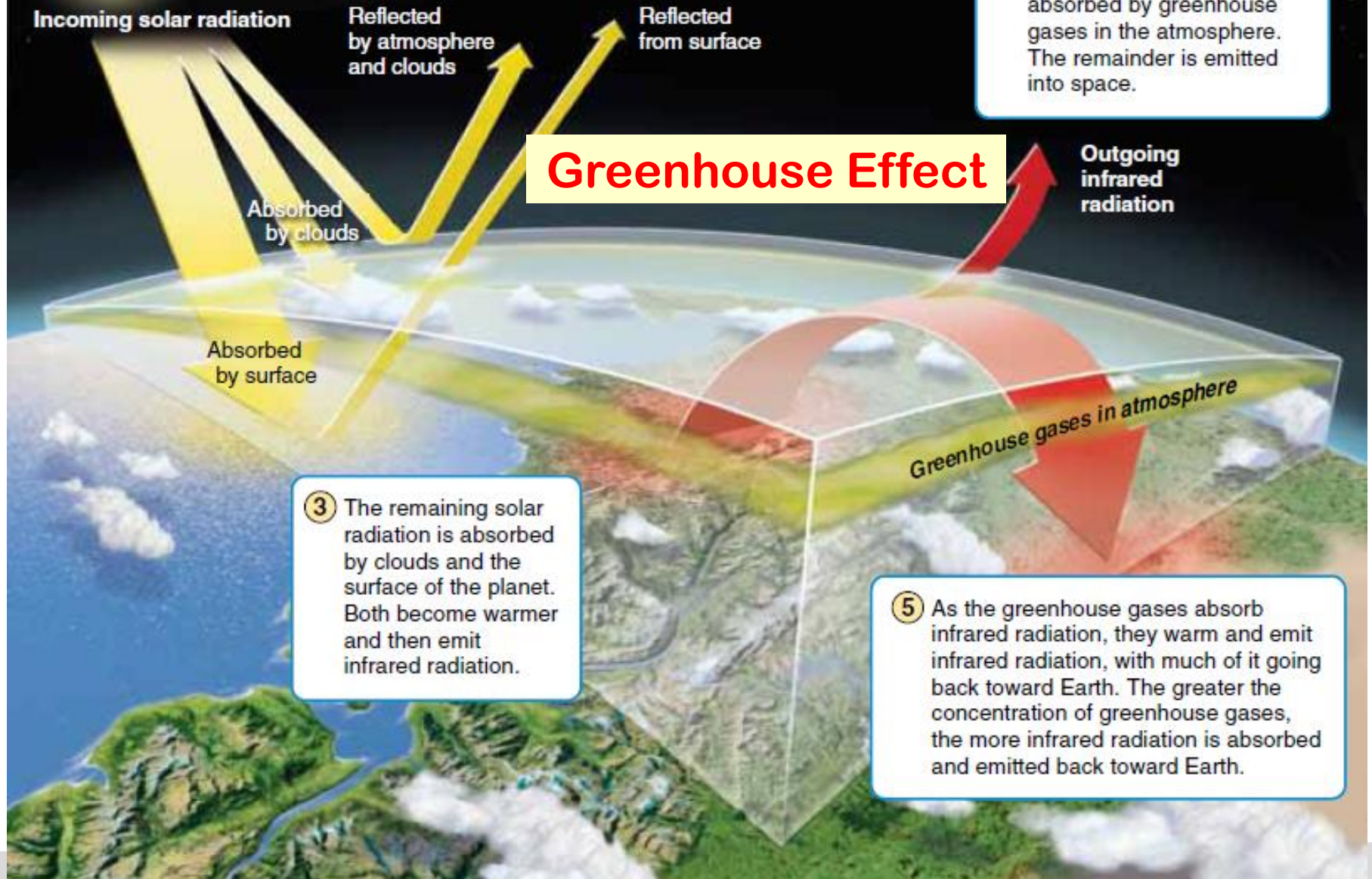
② About one-third of this solar radiation is reflected—from the atmosphere, clouds, and the surface of the planet—back into space.

④ Much of the emitted infrared radiation from Earth is absorbed by greenhouse gases in the atmosphere. The remainder is emitted into space.

③ The remaining solar radiation is absorbed by clouds and the surface of the planet. Both become warmer and then emit infrared radiation.

⑤ As the greenhouse gases absorb infrared radiation, they warm and emit infrared radiation, with much of it going back toward Earth. The greater the concentration of greenhouse gases, the more infrared radiation is absorbed and emitted back toward Earth.

Greenhouse Effect



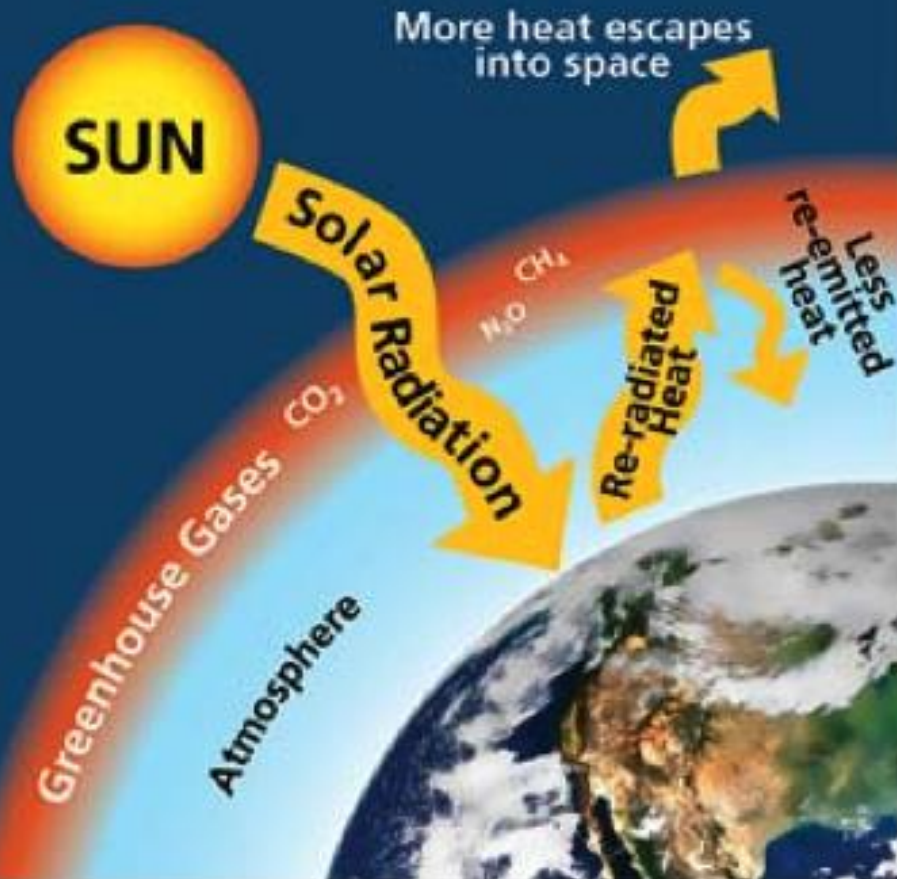
Greenhouse Gases

- Certain gases in the atmosphere can absorb infrared radiation emitted by the surface of earth and radiate much of it back toward the earth's surface. These gases are known as ***Greenhouse gases***.
- The most common gases in the atmosphere, N_2 and O_2 (99% of atmosphere) ***do not absorb*** IR radiation and are not greenhouse gases.
- ***Water vapour in the atmosphere absorbs more IR radiation than any other gas, but it do not persist as long as other greenhouse gases.***
- Other important greenhouse gases are Carbon dioxide (CO_2), Methane (CH_4), Nitrous oxide (N_2O) and Ozone (O_3).

Global Warming Potential

Greenhouse gas	Concentration in 2010	Global warming potential (over 100 years)	Duration in the atmosphere
Water vapor	Variable with temperature	<1	9 days
Carbon dioxide	390 ppm	1	Highly variable (ranging from years to hundreds of years)
Methane	1.8 ppm	25	12 years
Nitrous oxide	0.3 ppm	300	114 years
Chlorofluorocarbons	0.9 ppm	1,600 to 13,000	55 to >500 years

Natural Greenhouse Effect

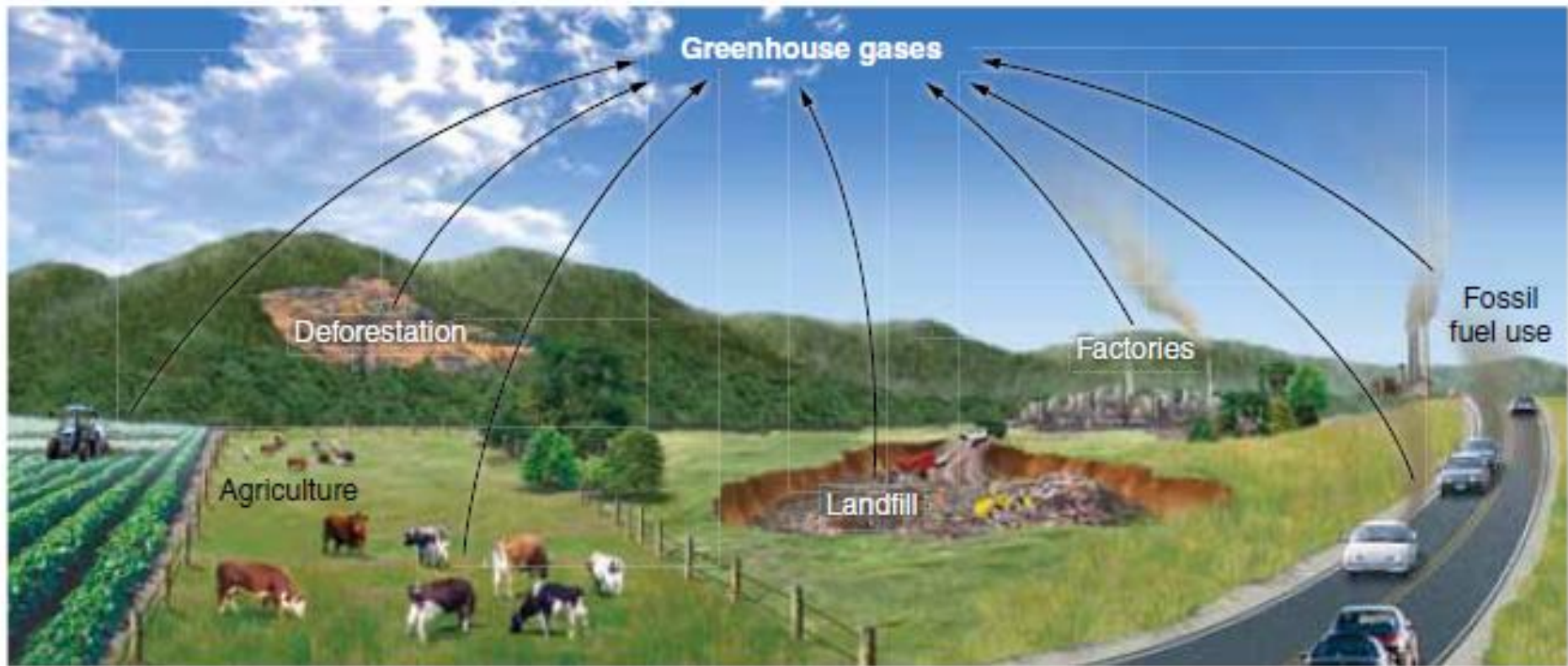


Human Enhanced Greenhouse Effect



Sources of Greenhouse Gases

- Anthropogenic sources of greenhouse gases are burning of fossil fuels, agriculture, deforestation, landfills and industrial production of chemicals.



Impact of Global Warming

- Rising Temperature
- Melting of poles
- Increased sea level
- Changes in water supply
- Damage to habitats
- Loss of biodiversity
- Floods and Droughts

TYPES OF BIOLOGICAL HAZARDS WE FACE:

A **non-transmissible disease** does not spread from one person to another but caused within the person's body. Such diseases tend to develop slowly and have multiple causes. Ex. cardiovascular diseases, most cancers, asthma, and diabetes.

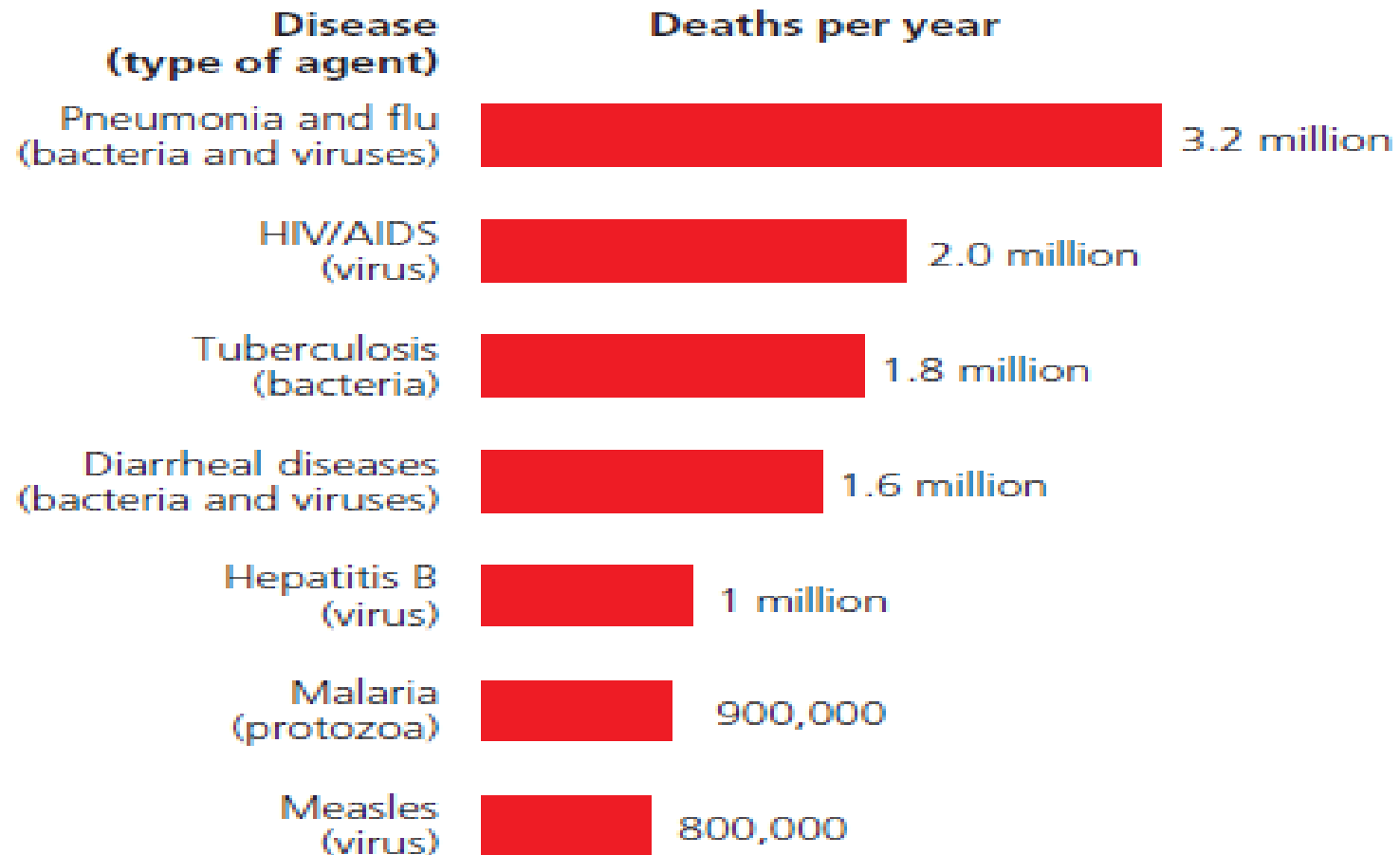
A **transmissible disease** (also called a *contagious* or *communicable disease*) is an infectious disease that can be transmitted from one person to another. Ex. TB, flu, HIV and measles. It is caused when a pathogen such as a bacterium, virus, or parasite invades the body and multiplies in its cells and tissues.

Pathways on which infectious disease organisms can enter the human body: through air, water, food, and body fluids (faeces, urine, blood), and mucus droplets sprayed by sneezing and coughing.

Epidemic: A large-scale outbreak of an infectious disease in an area or a country.

Pandemic: A global epidemic such as tuberculosis or AIDS.

Main infectious diseases which are deadly



In average about 11.3 million people every year killed by these diseases - equal to wiping out every one in Delhi